



JAIN COLLEGE OF ENGINEERING, BELAGAVI
Department of Electronics and Communication
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DOOR UNLOCKING SYSTEM
PROJECT GUIDE
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PRESENTED BY

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ABSTRACT

This project focuses on developing an IoT-based smart door lock system that combines RFID technology with the Blynk mobile application to enhance security and convenience. The system uses a NodeMCU (ESP8266) microcontroller connected to an RC522 RFID reader to identify authorized users through their RFID cards or tags. When a valid card is scanned, the system unlocks the door using a servo motor or relay mechanism, and the status of the lock is instantly updated on the Blynk app. The app also allows users to remotely monitor and control the door from anywhere via the internet. Unauthorized access attempts are denied and can trigger notifications for added safety. By integrating IoT connectivity with RFID-based authentication, this project offers a secure, low-cost, and user-friendly solution for modern smart homes and offices.."

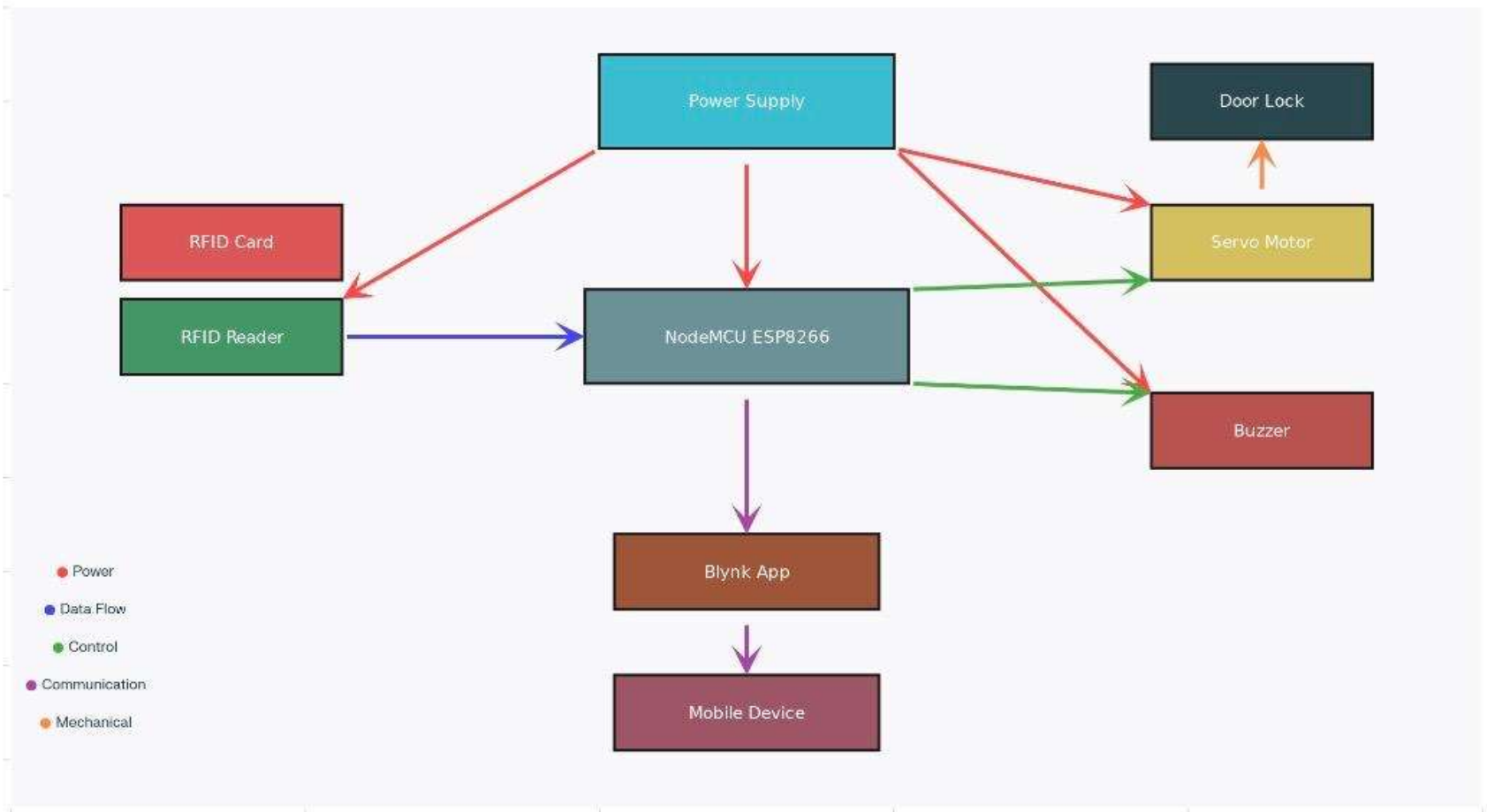
INTRODUCTION

- Security is a major concern in modern homes and offices.
- Traditional lock-and-key systems are less secure and inconvenient.
- This project implements an IoT-based smart door lock using RFID and the Blynk app.
- NodeMCU (ESP8266) controls the lock, and RC522 RFID reader verifies authorized cards.
- Door unlocks automatically via servo motor when a valid card is scanned.
- Blynk app allows remote monitoring and control of the door.
- Unauthorized access is denied and can trigger notifications.
- The system provides a secure, low-cost, and user-friendly smart access solution.

LITERATURE SURVEY

Reference Title with Author	Year	Technology	Results and Concusion
Al-kassab. design and implementation of RFID based security system.	2012	RFID technology likely including RFID reader,tags and possible microcontrollers.	Result:95% accuracy in authentication. Conclusion:the system was effective for security purpose
Chothia t. RFID authentication.	2019	RFID authentication protocols,likeky including encryption and mutual authentication method.	Result: they proposed and possible tested to improve security. Conclusion:emphasized the need foe better authentication to mitigate risks.
Finkenzeller k. Fundamental and application in contactless smartcards.	2003	RFID technology in general,covering tags,readers,freq and standards	Result:not applicable in the experimental sens,but it details the working of RFID. Conclusion:the book concludes with the broad applications and future of RFID.

RFID Door Unlock System



COMPONENTS REQUIRED

- NodeMCU (ESP8266): Main controller, connects to Wi-Fi, communicates with Blynk app.
- RC522 RFID Reader: Reads the unique ID from authorized RFID tags.
- Servo Motor : Mechanically unlocks or locks the door.
- Buzzer: Provides audio feedback for access granted, denied, or alerts.
- Blynk Mobile App: Allows remote monitoring and control of the door.
- Battery: power supply

ADVANTAGES

- ❖ Contactless operation.
- ❖ Fast and convenient access.
- ❖ Enhanced security.
- ❖ Easy to manage and update.
- ❖ Durable and low maintenance.
- ❖ Scalability.
- ❖ Customizable access control.
- ❖ Improved user experience.

DISADVANTAGES

- ❖ Loss or damage of tags.

APPLICATIONS

- ❖ office and corporate buildings.
- ❖ Educational institutions.
- ❖ Residential apartments and housing societies.
- ❖ Hospitals and laboratories.
- ❖ Industrial and factory premises.
- ❖ Hotels and hospitality sectors.
- ❖ Libraries and research center.
- ❖ Parking lots and vehicle gates.

REFERENCES

1. Muhammad Ali Mazidi, Janice Gillispie Mazidi, and Rolin D. McKinlay, The 8051 Microcontroller and Embedded Systems: Using Assembly and C, Pearson Education, 2008.→ Provides fundamental knowledge on microcontroller interfacing, often used for RFID-based systems.
2. Simon Monk, Programming Arduino: Getting Started with Sketches, McGraw-Hill Education, 2016.→ Explains how to develop embedded applications and control hardware like RFID readers via Arduino.
3. Atmel Corporation, ATmega328P Datasheet, Atmel, 2016.→ Technical documentation for the microcontroller used in Arduino Uno, including serial communication details relevant to RFID.
4. NXP Semiconductors, MFRC522 RFID Reader IC Datasheet, Rev. 3.7, NXP, 2019.→ The primary chip used in most low-cost RFID modules (13.56 MHz). Describes hardware connections and tag communication protocol.
5. ISO/IEC 14443, Identification cards — Contactless integrated circuit cards — Proximity cards, 2018.→ Defines communication standards for RFID systems operating at 13.56 MHz (used by MIFARE tags).

THANK YOU

